CHRIST (Deemed to be University)

Bangalore, India

Department of Electronics and Communication Engineering

Traffic Density Monitoring Using Drone Video and OpenCV

Project Report

Project Repository: https://github.com/pmappaiah-edu/Drone_Tech_Traffic_Monitoring

Submitted by:

PM Appiah – 2360647

Pothularaju Varshitha – 2360648

Qurrath Ul Ain - 2360649

Sanjana Krishna – 2360652

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Abstract :

Traffic congestion is a major issue in urban environments, leading to increased travel time, fuel consumption, and air pollution. Real-time traffic monitoring systems are essential for effective traffic management and smart city applications. This project presents a traffic density monitoring system using aerial drone video footage and computer vision techniques in OpenCV.

The system detects moving vehicles using the Background Subtraction (MOG2) algorithm and contour detection methods. Based on the number of detected vehicles per frame, traffic density is classified into Low, Medium, or High categories. The output is displayed in real time with bounding boxes around detected vehicles along with vehicle count and density level.

The proposed system provides an automated, scalable, and cost-effective solution for traffic monitoring and can be further enhanced using deep learning-based detection techniques.

1. Introduction

Traffic monitoring plays a crucial role in modern transportation systems. Traditional traffic monitoring methods such as manual counting, inductive loop sensors, and CCTV cameras have limitations in coverage and scalability.

With the advancement of drone technology and computer vision, aerial traffic monitoring has become an efficient alternative. Drone footage provides a wide field of view and better visibility of traffic patterns.

In this project, OpenCV is used to process drone video footage to:

- Detect moving vehicles
- Count vehicles per frame
- Classify traffic density levels
- Display real-time monitoring output

2. Aim

To develop a real-time traffic density monitoring system using drone video footage and OpenCV.

3. Objectives

- To detect vehicles using background subtraction.
- To apply image preprocessing techniques to reduce noise.
- To count the number of vehicles in each frame.
- To classify traffic density as Low, Medium, or High.
- To display real-time visual output.

4. Problem Statement

Urban authorities require an automated system to monitor traffic load using aerial images captured by drones. The system must:

1. Detect moving vehicles accurately.
2. Count vehicles in each video frame.
3. Classify traffic density levels.
4. Provide visual monitoring output in real time.

5. System Architecture

The system consists of the following stages:

1. Video Acquisition
2. Frame Preprocessing
3. Background Subtraction
4. Morphological Filtering
5. Contour Detection
6. Vehicle Counting
7. Density Classification
8. Output Display

6. Methodology

6.1 Video Acquisition

A pre-recorded drone video is used as input. The video is read frame-by-frame using OpenCV's Video Capture function.

6.2 Frame Preprocessing

Each frame is:

- Resized to 800 × 600 pixels for uniform processing.
- Gaussian Blur is applied to reduce noise and smooth the image.

Gaussian Blur helps in reducing small variations in pixel intensity, improving detection accuracy.

6.3 Background Subtraction

The MOG2 (Mixture of Gaussians) algorithm is used to separate moving objects from the background.

Parameters used:

- History = 1000
- Variance Threshold = 25
- Shadow Detection = Enabled

The output is a foreground mask highlighting moving vehicles.

6.4 Morphological Operations

To remove noise and improve object detection:

- Morphological Opening removes small unwanted noise.
- Morphological Closing fills small gaps inside detected objects.

A 5 × 5 kernel is used for filtering.

6.5 Contour Detection

Contours are extracted from the foreground mask using:

- cv2.findContours()

Each contour area is calculated.
If contour area > 400 pixels, it is considered a vehicle.

Bounding rectangles are drawn around detected vehicles.

6.6 Vehicle Counting

Each valid contour increments the vehicle counter.

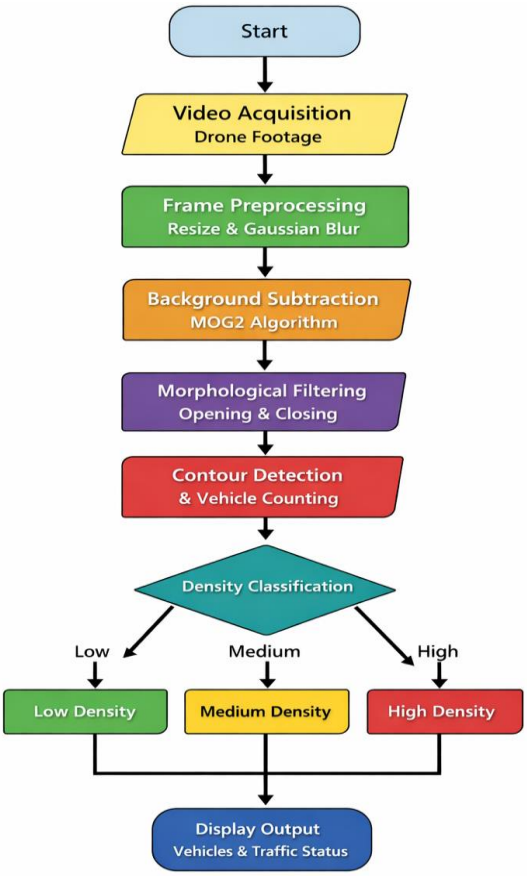
Vehicle Count = Number of contours with area greater than threshold.

6.7 Density Classification

Traffic density is classified based on vehicle count:

Vehicle Count	Density Level	Display Colour
< 10	LOW	Green
10 – 24	MEDIUM	Yellow
≥ 25	HIGH	Red

7. Algorithm



8. Implementation

The project is implemented using:

- Python Programming Language
- OpenCV Library
- NumPy Library

The system runs in real time and displays:

- Detected vehicles with bounding boxes
- Vehicle count
- Traffic density classification

9.Results

9.1 Low Traffic Condition

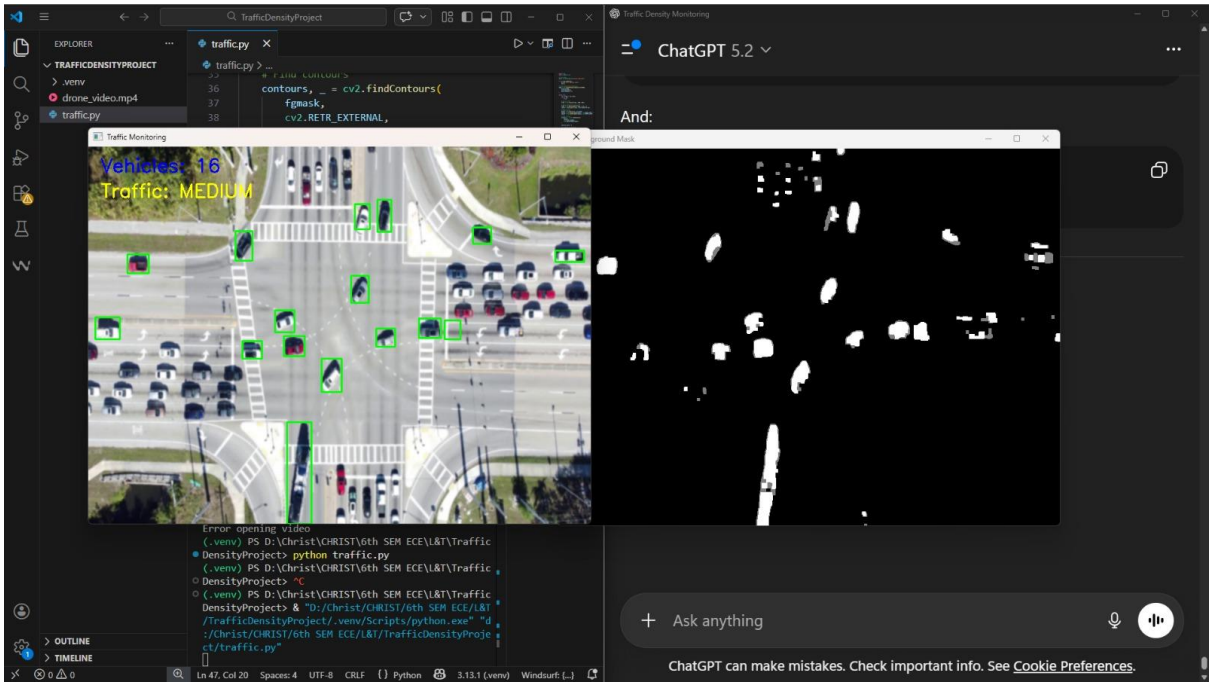
Observation:

- Vehicle Count ≈ 8
- Classified Density: LOW
- Bounding boxes correctly detected moving vehicles.
- Foreground mask shows clear separation of moving objects.

9.2 Medium Traffic Condition

Observation:

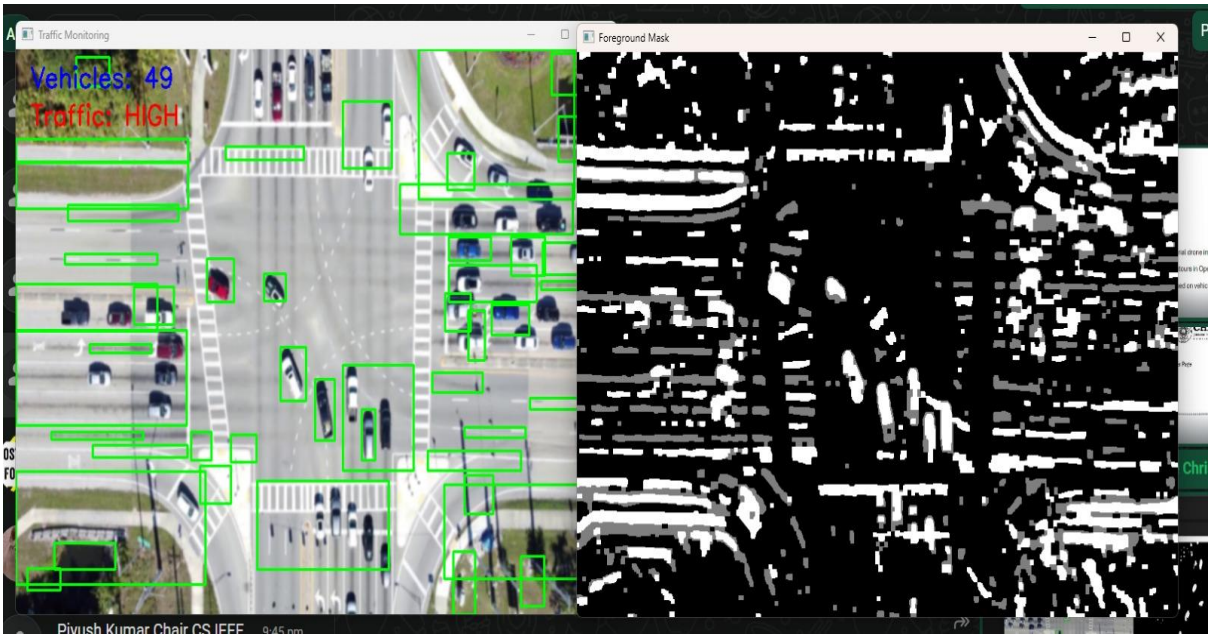
- Vehicle Count ≈ 16
- Classified Density: **MEDIUM**
- Most vehicles are correctly bounded.
- Minor noise present in the foreground mask.



9.3 High Traffic Condition

Observation:

- Vehicle Count ≈ 49
- Classified Density: **HIGH**
- Multiple bounding boxes detected across lanes.
- Foreground mask shows dense white regions.



10. Applications

- Smart Traffic Management Systems
- Urban Planning and Analysis
- Emergency Route Optimization
- Smart City Infrastructure
- Real-Time Congestion Monitoring

11. Advantages

- Automated monitoring
- Low-cost implementation
- Scalable solution
- Real-time analysis
- Easy integration with smart city systems

12. Limitations

- Sensitive to lighting changes
- Shadows may affect detection accuracy
- Stationary vehicles may not be detected
- Performance may reduce during heavy rain or fog

13. Future Scope

- Integration of Deep Learning models such as YOLO for better accuracy
- Real-time live drone feed processing
- Database logging of traffic statistics
- Cloud-based monitoring system
- Integration with traffic signal control systems

14. Conclusion

This project demonstrates a real-time traffic density monitoring system using drone video footage and OpenCV. By applying background subtraction and contour detection techniques, moving vehicles are detected and counted effectively. The system classifies traffic density levels and provides real-time visual feedback.

The proposed approach offers a scalable and automated solution for urban traffic monitoring. With further enhancements using deep learning and cloud integration, the system can be deployed in smart city environments for intelligent traffic management.